Milton Tran

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Education

University of California, Riverside — Riverside, CA
Undergraduate Student in Mechanical Engineering — *Expected Graduation: June 2030*

Engineering Experience

FRC Team 9470 — Fremont, CA
Mechanical Design Lead — *June 2024 – May 2026*

- Led mechanical design and integration, defining robot architecture and coordinating subsystem layouts to support overall performance and serviceability under one week deadlines.
- Designed a custom elevator mechanism achieving 5 ft of vertical travel in under 0.8 s to rapidly position PVC pipes for scoring.
- Designed a funnel mechanism to index PVC pipes into an end effector in under 1 second
- Designed and optimized a shooting mechanism capable of emptying a hopper containing 60+ game pieces in under 3 seconds.
- Coordinated pit operations, maintenance, and mechanical troubleshooting during competitions to restore robot functionality in under ten minutes.
- Collaborated with FRC Teams 1678 and 254 at district and world championship events to exchange engineering ideas and improve robot performance.

Highlander Racing — Riverside, CA
Suspension Subteam — *September 2026 – Present*

- Designed SolidWorks suspension components, including 4130 A-arm assemblies and mounting brackets, integrating rod ends, spherical bearings, and packaging constraints.
- Performed FEA and analytical buckling analysis on 10.75-inch, 4130 steel suspension tubes, evaluating loads up to 900 lbf and comparing simulation results against hand calculations.
- Conducted structural tear-out analysis on 0.1-inch-thick mounting brackets, finding a 16.18 kN ultimate tear-out load with a factor of safety of 2 and evaluating suspension loading using tire forces up to 3.5 kN and moments up to 60 N·m, verifying the design.
- Documented suspension component designs, material selections, analytical calculations, and FEA results in a technical report, including buckling and tear-out analyses to support design validation and future iterations.

Highlander Combat Robotics — Riverside, CA
Mechanical Subteam — *October 2026 – Present*

- Design mechanical components and assemblies in Onshape for a plastic combat robot, limited to PLA, PETG and ABS, accounting for dimensional constraints, component interfaces, and manufacturability.
- Designed using parametric design practices to make future edits or architecture changes smoother
- Created and maintained spreadsheets to track component weights and dimensions, establishing a design budget to manage packaging constraints and meet the 454 gram competition weight limit as well as run tear out analysis and motor speed calculations.
- Documented the end-to-end design and build process, recording CAD iterations, component specifications, slicer setting decisions, and assembly procedures to improve design traceability and team collaboration.

Awards

FIRST World Championship: 2026 Archimedes Division Champion
FIRST District Championships: 2026 Half Moon Bay and Silicon Valley District Champion
Engineering Awards: FIRST Excellence in Engineering and Industrial Design Awards

Technical Skills

CAD and Design Software: Onshape, SolidWorks, Fusion 360, OrcaSlicer
Manufacturing and Fabrication: FDM and SLA 3D Printing, CNC Machining, Assembly
Electronics: Arduino Uno, Universal G-code Sender, Wiring
Programming and Tools: Python, Java, VS Code